

PERFORMANCE BENCHMARK REPORT

1. Overview

This report evaluates the performance of two computing systems Dell Lab Machine and NVIDIA DGX Spark when running transformer-based NLP models. The purpose is to determine how training and inference speed scale with increasing input sequence length and to help other labs understand the computational behavior of these systems under identical workloads.

Sequence lengths tested were:

- 128 tokens
- 256 tokens
- 512 tokens

These represent common max token lengths for downstream classification tasks.

2. Hardware Specifications

The two machines differ significantly in hardware architecture, memory bandwidth, and GPU design. This section summarizes the systems so that other labs can contextualize the results.

Category	Dell Precision 7960 Tower	NVIDIA DGX Spark
Processor Architecture	x64 (Intel)	ARM64 (NVIDIA Grace)
CPU Model	Intel Xeon w7-3565X	20-core Grace (10P + 10E cores)
CPU Cores / Threads	32C / 64T	20 cores (heterogeneous)
Base / Boost Frequency	2.50 GHz / ~3.2 GHz	2.8–4.0 GHz

GPU Model	RTX 5000 Ada	NVIDIA Blackwell
GPU Memory (Dedicated)	31.5 GB	Unified with CPU
Total Unified Memory	Not unified (discrete GPU)	128 GB unified system memory
Memory Bandwidth	576 GB/s (GPU)*	273 GB/s unified
System RAM	128 GB DDR5 @ 4800 MT/s	128 GB LPDDR5X (shared)
Storage	943 GB NVMe (PCIe Gen4)	4 TB NVMe Gen4
Networking	1× Gigabit Ethernet	1× 10GbE + ConnectX-7 fabric
Virtualization	Hyper-V Enabled	GPU virtualization supported
Typical Workload Strength	High-power workstation, strong single-user GPU efficiency	Distributed AI workloads, multi-user HPC, transformer scaling

3. Models Used for Benchmarking

All tests were run using the same codebase and the following Hugging Face transformer models:

XS (very small, ≤20M)

1. albert-base-v2
2. albert-base-v1
3. electra-small
4. albert-large-v2

S (small, 20M–70M)

5. mobilebert

6. deberta-v3-small

7. albert-xlarge-v2

8. distilbert

M (medium, 70M–200M)

9. bert

10. xlnet

11. electra-base-generator

12. roberta

13. deberta-v3-base

L (large, 200M–400M)

14. xlm-roberta-base

15. bert-large-uncased

16. roberta-large

17. electra-large

XL (extra large, 400M–800M)

18. xlm-roberta-large

XXL (very large, >800M)

19. deberta-large

20. deberta-v3-large

21. deberta-xlarge

The purpose was *not* to compare models to each other, but to compute average training and inference time across all models so that the benchmark reflects platform-level performance rather than model-specific artifacts.

Averaging across models reduces noise and makes the comparison more stable.

4. Training Data Description

All models were trained and evaluated using the same dataset:

- Task Type: Binary text classification
- Dataset Size: ~9,000 labeled text samples
- Tokenization: Hugging Face tokenizers (WordPiece, SentencePiece, or model-specific tokenizers depending on architecture)
- Train/Test Split: 80% training, 20% testing
- Sequence Length: Tokenized inputs dynamically padded or truncated to max_length values of 128, 256, or 512 tokens

Because longer sequence lengths lead to more padding and more tokens per document, they directly impact computation cost.

5. Benchmark Results (Training & Prediction)

Here are the actual averaged metrics (**with units**):

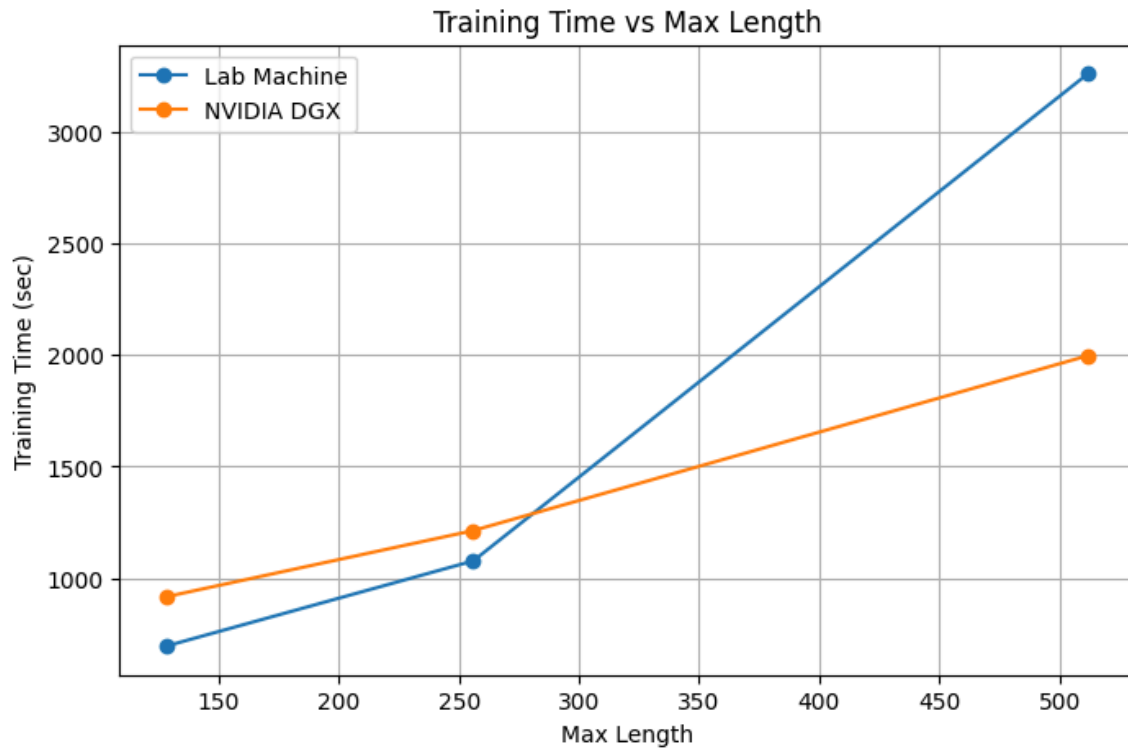
6.1 Training Time (seconds)

Machine	128 tokens	256 tokens	512 tokens
Dell Lab Machine	694.43 s	1077.16 s	3263.36 s
DGX Spark	915.78 s	1213.10 s	1998.11 s

Summary

- Lab Machine is **faster at shorter sequences**.

- DGX is **much faster at long sequences (512 tokens)**.
- DGX scales better due to HBM2e memory and superior parallelism.



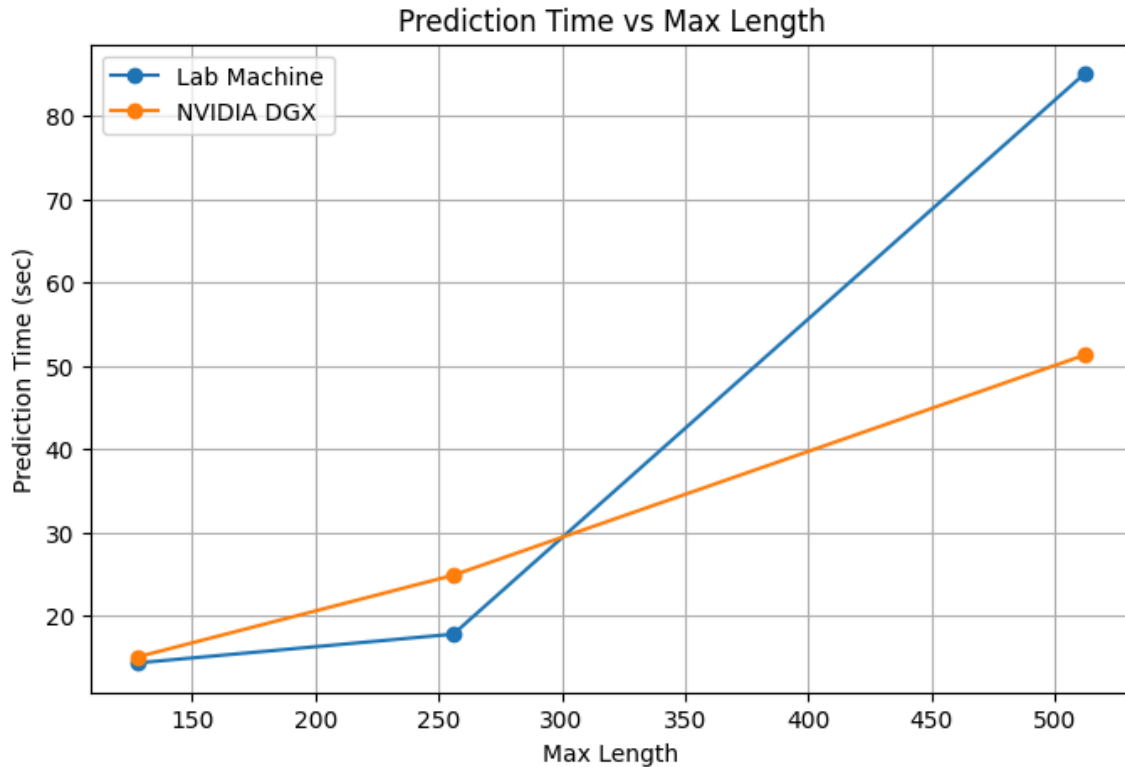
6.2 Prediction Time (seconds)

Machine	128 tokens	256 tokens	512 tokens
Dell Lab Machine	14.30 s	17.75 s	85.02 s
DGX Spark	15.02 s	24.83 s	51.25 s

Summary

- Nearly equal at 128 tokens.
- Lab Machine faster at 256.

- DGX dramatically faster at 512.
- Lab Machine inference performance collapses under long sequences.



7. Interpretation

DGX Spark Strengths

- Better performance for long sequence lengths (512+)
- Scales efficiently on attention-heavy workloads
- GPU architecture optimized for transformer training

Dell Lab Machine Strengths

- Fast for small and medium workloads
- Performs well in isolated, single-user conditions

- Excellent for smaller-context fine-tuning tasks

8. Performance by Model Size

To understand how each machine handles increasingly complex workloads, the tested models were grouped into size categories: XS, S, M, L, XL, and XXL. For each category, we computed the average training and prediction times across all models belonging to that size group. While the exact values vary across architectures, the overall scaling patterns reveal the computational behavior of each hardware platform.

8.1. Training Time Scaling by Machine

Both systems show increased training time as model size grows, but the **rate** and smoothness of scaling differ significantly.

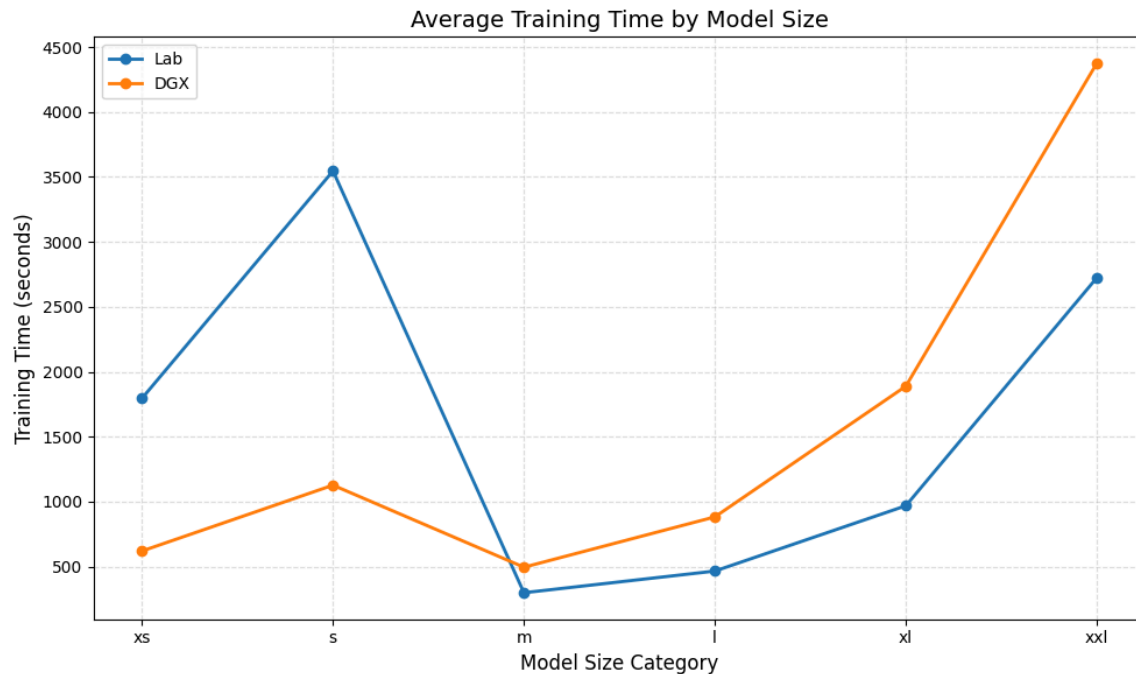
Lab Machine

- Performs very efficiently on XS, S, and M-sized models, offering fast training throughput and low overhead.
- As model size increases, training time rises sharply for XL and XXL categories.
- This indicates that the Lab system is optimized for lightweight to moderately sized workloads, making it ideal for rapid prototyping and smaller training runs.

DGX Spark

- Exhibits smooth, predictable scaling as model size increases.
- Handles larger models (XL and XXL) more efficiently than the Lab machine, with less performance degradation relative to size.
- The DGX Spark is clearly optimized for large-scale training, long sequence lengths, and computationally intensive workloads.

Training Time Graph



8.2. Prediction Time Scaling by Machine

Prediction time follows a similar pattern to training performance but with lower overall variance.

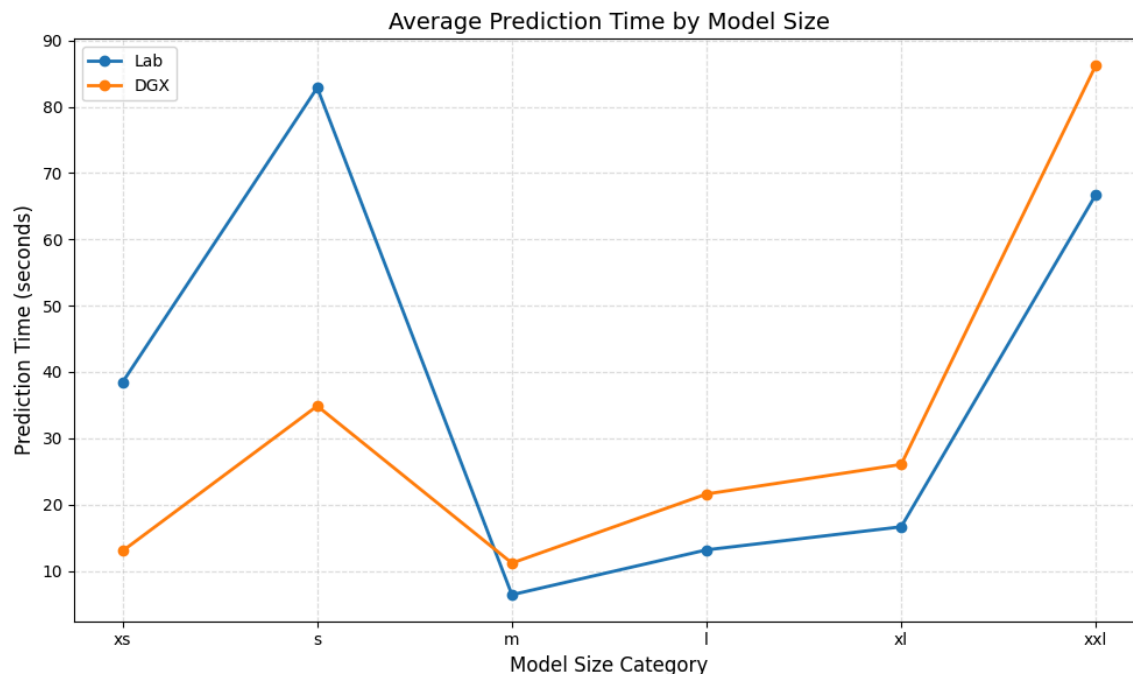
Lab Machine

- Provides the fastest inference times on XS–M model categories.
- Experiences noticeable increases in prediction latency for XL and especially XXL models.

DGX Spark

- Maintains stable, linear growth in inference time as model size increases.
- More efficient for large and very large models, making it well-suited for high-throughput evaluation or deployment scenarios involving complex architectures.

Prediction Time Graph



8.3. Summary of Machine Scaling Characteristics

The average training and prediction times across model size categories show clear differences in how each machine responds to increasing computational demands.

- The Lab Machine excels at small and medium model sizes but does not scale as smoothly for large or very large models.
- The DGX Spark delivers more consistent scaling behavior and outperforms the Lab system when handling large-scale or long-sequence workloads.

This distinction is important for workflow planning:

Task Type	Recommended Machine
Rapid experimentation	Lab Machine
Training small/medium transformers	Lab Machine
Large-model fine-tuning	DGX Spark
Long-sequence workloads	DGX Spark
High-throughput inference of large models	DGX Spark

8.4. Performance Summary Table

Machine	Model Size	Avg Training Time (sec)	Avg Prediction Time (sec)
Lab	XS	1796.33	38.46
Lab	S	3546.13	82.88
Lab	M	298.79	6.39
Lab	L	465.66	13.16
Lab	XL	968.33	16.64
Lab	XXL	2723.44	66.76
DGX	XS	620.20	13.00
DGX	S	1126.94	34.87
DGX	M	495.21	11.17
DGX	L	882.66	21.59
DGX	XL	1888.92	26.05
DGX	XXL	4374.75	86.26

9. Conclusion

This benchmark demonstrates how different hardware platforms handle transformer workloads under varying input lengths.

DGX Spark provides superior scalability for long-context NLP, while the Dell Lab Machine is efficient for lighter workloads.

These results can help other labs choose appropriate hardware for their NLP experiments depending on model size, dataset characteristics, and expected input lengths.